

BSc (Hons) in Computer Science
BSc (Hons) in Computer Systems Engineering
Year 2

Group Assignment

SE2012 – Object Oriented Design and Analysis

Semester 2, 2026

Release Date: 07/08/2026

General Instructions:

1. This assignment carries **20% out of final marks**.
2. This assignment must be done in groups of 4 members from the same group.
3. Late submissions will be penalized.
4. The submission file should be a PDF document.
5. The Taks deadlines and submission link will be published on the course website.

Each group should do the following.

1. Identify a **real-world client** matching for the given Criteria. Usually, any SME would be sufficient.
 - a. **Submit the project details.**
 - i. Topic
 - ii. Client details
 - iii. Each member's scope
 1. Each member must have equal scope which **covers all the aspects of development**.
 2. This should include both Front-end and back-end.
 3. **No member** should focus only on one aspect of development. (that is Front-end or back-end)
2. **Gather requirements from the client.**
 - a. Write a project overview.
 - b. Identify all functional requirements for the system.
 - c. Identify all non-functional requirements for the system.
 - d. Identify at least 4 main functions (one for each member). All the functions should be approximately equal in scope and contain a considerable amount of work.
 - e. Gather all details related to each function.
 - f. Develop the UML Use-case diagram for the system.
 - g. Write the Use-case scenarios for 4 main use-cases. (one for each member)
 - h. Develop Activity diagrams for the identified use-cases.

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- i. Develop a UML Class diagram.
- j. Compile a report according to the given template.

3. Implement the solution using Java Spring Boot as a Spring Web Application

- a. Develop the UI for the front-end Web application
- b. Backend (with Restful API and Spring MVC)
- c. A basic tutorial could be found here [Tutorial: Create your first Spring application | IntelliJ IDEA Documentation \(jetbrains.com\)](#)

Submission Requirements

- Students must use the GitHub Teams Repository assigned to them for coding. Individual contributions will be assessed based on the GitHub commits and activities.
- **Project Details (25th August 2026) – No Marks**
- **Initial Presentation – 2nd September (During Practical) - 2.5%**
- **Progress evaluation – 30th September (During Practical) – 5%**
- **Group Report.** (to be submitted by the **07 October 2026** to Turnitin) – **2.5%**
- **Viva and the code submission (Gradescope).** (**21 October 2026**) - **10%**

[Grading Criteria](#)

Initial Presentation (2.5%)

Section	Description	Marks
Client Details	Group Marks	10
Identified Stakeholders/Actors		20
Stakeholder meetings to gather requirements		20
Identified Scope/Features		20
Individual Responsibilities		20
Communication		10

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Progress Presentation (5%)

Section	Description	Marks
Finalized Requirements	Group	10
Use cases	Individual	20
Identified Tasks for Individual Components	Individual	20
Initial UIs and Backend Implementation (50% completion)	Individual	20
Deployment Plan	Group	20
Communication and Presentation	Individual	10

Group Report (2.5%)

Section	Description	Marks
Cover page		5 marks
Table of contents		
List of figures		
List of tables		
Introduction	Client details, Project Scope, Overview	5 marks
Functional Requirements		10 marks
Non-functional Requirements		10 marks
Use case diagram		15 marks
Use case scenarios (4)	One per student Minimum	15 marks
Activity diagrams (4)	One per student Minimum	20 marks
Class diagram		15 marks
Appendix	Evidence of client meetings or related documents	5 marks

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Final VIVA and Product (10%)

Criteria	Description	Marks
OOO and Design Patterns	Individual Mark	20
Best Practices	Individual Mark	10
Members' Contribution to the Group	Individual Mark	15
Testing	Individual Mark	10
Deployment	Group/Individual	10
Presentation Skills	Individual Mark	10
Demo	Individual Mark	25

Weighted Group Marks

- Please note all Group marks will be weighted based on the peer feedback from the group members.
- Please inform any conflicts or issues faced as soon as possible. Do not bring the concerns on the last week of the Semester.